

Math 3IA3 - Assignment 1

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Exercise 1.1.15

Show that if $f : A \rightarrow B$ and G and H are both subsets of B , then:

(a) $f^{-1}(G \cup H) = f^{-1}(G) \cup f^{-1}(H).$

Solution:

We will prove that $f^{-1}(G \cup H) = f^{-1}(G) \cup f^{-1}(H)$ by first showing that $f^{-1}(G) \cup f^{-1}(H) \subseteq f^{-1}(G \cup H)$ and that $f^{-1}(G \cup H) \subseteq f^{-1}(G) \cup f^{-1}(H).$

(I) Show that $f^{-1}(G) \cup f^{-1}(H) \subseteq f^{-1}(G \cup H).$

Let $b \in (f^{-1}(G) \cup f^{-1}(H)).$ Then, by definition of inverse, we know $f(b) \in G \vee f(b) \in H.$ Thus, $f(b) \in G \cup H.$ Taking the inverse of both sides, yields $b \in f^{-1}(G \cup H).$ Thus, since $b \in (f^{-1}(G) \cup f^{-1}(H)) \implies b \in f^{-1}(G \cup H),$ $(f^{-1}(G) \cup f^{-1}(H)) \subseteq f^{-1}(G \cup H).$

(II) Show that $f^{-1}(G \cup H) \subseteq f^{-1}(G) \cup f^{-1}(H).$

Let $b \in f^{-1}(G \cup H).$ Applying f to both sides yields $f(b) \in (G \cup H) \implies f(b) \in G \vee f(b) \in H.$ Thus, reapplying the inverse we have that $b \in f^{-1}(G) \vee b \in f^{-1}(H),$ which, by definition of inverse implies that $b \in f^{-1}(G) \cup f^{-1}(H).$ Thus, since $b \in f^{-1}(G \cup H) \implies b \in (f^{-1}(G) \cup f^{-1}(H)),$ $f^{-1}(G \cup H) \subseteq (f^{-1}(G) \cup f^{-1}(H)).$

Therefore, since $f^{-1}(G) \cup f^{-1}(H) \subseteq f^{-1}(G \cup H)$ and $f^{-1}(G \cup H) \subseteq f^{-1}(G) \cup f^{-1}(H),$ we necessarily have that $f^{-1}(G \cup H) = f^{-1}(G) \cup f^{-1}(H).$

(b) $f^{-1}(G \cap H) = f^{-1}(G) \cap f^{-1}(H).$

Solution:

We will likewise prove that $f^{-1}(G \cap H) = f^{-1}(G) \cap f^{-1}(H)$ by first showing that $f^{-1}(G \cap H) \subseteq (f^{-1}(G) \cap f^{-1}(H))$ and that $(f^{-1}(G) \cap f^{-1}(H)) \subseteq f^{-1}(G \cap H).$

(I) Show that $f^{-1}(G \cap H) \subseteq (f^{-1}(G) \cap f^{-1}(H)).$

Let $b \in f^{-1}(G \cap H).$ Therefore, applying the function to both sides, we know that $f(b) \in (G \cap H) \implies f(b) \in G \wedge f(b) \in H.$ Thus, applying the inverse, we have $b \in f^{-1}(G) \wedge b \in f^{-1}(H)$ and thus, by definition of intersection, we have $b \in f^{-1}(G) \cap f^{-1}(H).$ Thus, since $b \in f^{-1}(G \cap H) \implies b \in (f^{-1}(G) \cap f^{-1}(H)),$ $f^{-1}(G \cap H) \subseteq (f^{-1}(G) \cap f^{-1}(H)).$

(II) $(f^{-1}(G) \cap f^{-1}(H)) \subseteq f^{-1}(G \cap H).$

Let $b \in f^{-1}(G) \cap f^{-1}(H).$ Therefore, applying the function to both sides, we know that $f(b) \in G \wedge f(b) \in H.$ Thus, by definition of inverse, we know that $f(b) \in (G \cap H),$ and by reapplying the inverse to both sides, we yield $b \in f^{-1}(G \cap H).$ Thus, since

$$b \in (f^{-1}(G) \cap f^{-1}(H)) \implies b \in f^{-1}(G \cap H), (f^{-1}(G) \cap f^{-1}(H)) \subseteq f^{-1}(G \cap H).$$

Therefore, since $f^{-1}(G \cap H) \subseteq (f^{-1}(G) \cap f^{-1}(H))$ and $(f^{-1}(G) \cap f^{-1}(H)) \subseteq f^{-1}(G \cap H)$ we necessarily have that $(f^{-1}(G) \cap f^{-1}(H)) = f^{-1}(G \cap H)$

Exercise 1.2.16

Conjecture a formula for the sum of the first n odd natural numbers $1 + 3 + \dots + (2n - 1)$ and prove your formula using mathematical induction.

Solution:

First n odd natural numbers	Sum of first n odd natural numbers
1	$1 = 1$
2	$1 + 3 = 4$
3	$1 + 3 + 5 = 9$
4	$1 + 3 + 5 + 7 = 16$
5	$1 + 3 + 5 + 7 + 9 = 25$

This table leads me to conjecture that the sum of the first n odd natural numbers is equal to n^2 . To attempt to prove this, I will use mathematical induction. Let $P(n)$ be the statement $P(n) : 1 + 3 + \dots + (2n - 1) = n^2$.

(I) Prove the base case (Show $P(1)$ is true).

LHS:	RHS:
$= 1$	$= 1^2$
$= 1$	$= 1$

Therefore, since LHS = RHS, $P(1)$ holds.

(II) State the induction hypothesis (State $P(k)$).

Let $k \in \mathbb{N}$ be arbitrary. We assume, for the purposes of our induction hypothesis that the statement:

$$P(k) : 1 + 2 + \dots + (2k - 1) = k^2$$

is true.

(III) Prove the induction step (Prove $P(k) \implies P(k + 1)$).

$$\begin{aligned}
 1 + 2 + \dots + (2k - 1) + (2k - 1) &= k^2 + (2k + 1) \\
 &= (k + 1)^2
 \end{aligned}$$

Thus, since $P(1)$ holds true, and for arbitrary $k \in \mathbb{N}$, $P(k) \implies P(k+1)$, by the principle of mathematical induction, $P(n)$ holds $\forall n \in \mathbb{N}$.

Exercise 2.1.6

Use the argument in the proof of Theorem 2.1.4 to show that there does not exist a rational number s such that $s^2 = 6$.

Solution:

Let's assume, for the purposes of deriving a contradiction, that such a rational number s exists. We can therefore express s in lowest form $\frac{p}{q}$ where p and q are coprime. We can therefore surmise that:

$$\begin{aligned}\left(\frac{p}{q}\right)^2 &= 6 \\ p^2 &= 6q^2\end{aligned}$$

Therefore, we can express p^2 as a multiple of $6q^2$ and thus p^2 must be an even number, since 6 is trivially divisible by 2. Since p^2 is even, we therefore know p must be even as well. Thus, we can express p as $2k$, where $k \in \mathbb{Z}$. Following this, we find that:

$$\begin{aligned}\frac{p^2}{q^2} &= 6 \\ \frac{(2k)^2}{q^2} &= 6 \\ \frac{4k^2}{q^2} &= 6 \\ 4k^2 &= 6q^2 \\ 2k^2 &= 3q^2\end{aligned}$$

Since $2k^2$ is necessarily even, this implies that $3q^2$ must also be even. However, for $3q^2$ to be even, that must mean that q^2 is even since 3 is not an even number. Thus, if q^2 is even then so must q . However, this presents a contradiction to our earlier assumption that p and q are coprime, since we have just deduced that both p and q are both divisible by 2. Therefore, by contradiction, there cannot exist a rational number s such that $s^2 = 6$.

Exercise 2.3.6

Let S be a nonempty subset of $\mathbb{R}$ that is bounded below. Prove that $\inf S = \sup \{-s : s \in S\}$.

Solution:

Proof. Appealing to the completeness property of $\mathbb{R}$, we can assert that $\inf S$ must exist since it is bounded below. Thus, let $m = \inf S$. By definition of the infimum, we know that $\forall s \in S, m \leq s$ and therefore $-m \geq -s$. Therefore, we know that $-m$ must upperbound the set $\{-s : s \in S\}$.

To prove that $-m$ must be the least upperbound, and thus the supremum, of $\{-s : s \in S\}$, consider that for arbitrary $\epsilon > 0$, $\exists s_0 \in S$ such that $m + \epsilon > s_0$. Therefore, $-m - \epsilon < -s_0$, meaning, since $\epsilon > 0$ was arbitrary that $-m = \sup \{-s : s \in S\}$. Therefore, deferring to our prior definition of m , we can see that:

$$\begin{aligned} m &= m \\ m &= -(-m) \\ \inf S &= -(\sup \{-s : s \in S\}) \\ \inf S &= -\sup \{-s : s \in S\} \end{aligned}$$

□

Exercise 2.5.12

Give the two binary representations of:

(a) $\frac{3}{8}$

Solution:

Since $\frac{3}{8} < \frac{1}{2}$, we take a_1 to be equal to 0 and therefore we have:

$$\frac{3}{8} \in \left[0, \frac{1}{2}\right]$$

Now, since $\frac{3}{8} > \frac{1}{4}$, we can take a_2 to be equal to 1, thus giving:

$$\frac{3}{8} \in \left[\frac{1}{4}, \frac{1}{2}\right]$$

However, now the point of bisection of the interval $\left[\frac{1}{4}, \frac{1}{2}\right]$ is given by $\frac{1}{4} + \frac{\frac{1}{2} - \frac{1}{4}}{2} = \frac{3}{8}$. Therefore, we can express $\frac{3}{8}$ in binary form as:

$$\frac{3}{8} = (0.011000\dots)_2$$

or

$$\frac{3}{8} = (0.010111\dots)_2$$

(b) $\frac{7}{16}$

Solution:

Since $\frac{7}{16} < \frac{1}{2}$, we take a_1 to be equal to 0 and therefore we have:

$$\frac{7}{16} \in \left[0, \frac{1}{2}\right]$$

Now, since $\frac{7}{16} > \frac{1}{4}$, we can take a_2 to be equal to 1, thus giving:

$$\frac{7}{16} \in \left[\frac{1}{4}, \frac{1}{2}\right]$$

Since $\frac{7}{16} > \frac{3}{8}$, we set a_3 equal to 1 and attain:

$$\frac{7}{16} \in \left[\frac{3}{8}, \frac{1}{2} \right]$$

However, now the point of bisection of the interval $\left[\frac{3}{8}, \frac{1}{2} \right]$ is given by $\frac{3}{8} + \frac{\frac{1}{2} - \frac{3}{8}}{2} = \frac{7}{16}$. Therefore, we can express $\frac{7}{16}$ in binary form as:

$$\frac{7}{16} = (0.0111000\dots)_2$$

or

$$\frac{7}{16} = (0.0110111\dots)_2$$